MANCHESTER 1824
The University of Manchester

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Embedding skills for a new profession by teaching programming in an immersive and authentic environment

1 Clinical Bioinformatics – Our Students

Clinical Bioinformatics is a relatively new profession bringing together disciplines of Computer Science and Genomics within clinical practice. Those new to the field come from many different backgrounds; be it people from the health sector with little or no programming experience to those with IT knowledge but with limited clinical experience

Distance Learners
This postgraduate course was delivered fully online to students in full time roles juggling commitments in both work and home life. Therefore it was essential to build an open and supportive learning environment to encourage the students to seek help when they needed it. We wanted to give them time to practice the coding activities at a pace to suit their own ability & contexts whilst ensuring momentum was maintained for the cohort as a whole.

Community of Practice (CoP)
Helping to build a Clinical Bioinformatics CoP has been an essential theme running through our teaching for a long time[1]. This practical course needed to include this aspect and to emulate a professional network. This was important not only during the course but also post-course so they continue to share knowledge and develop their programming skills back in practice.

1. Davies AC, Harris D, Banks-Gatenby A, Brass A. Problem-based learning in clinical bioinformatics education: Does it help to create communities of practice?. PLoS Comput Biol. 2019;15(6):e1006746. Published 2019 Jun 27. 6

2 Authentic & Immersive environment

Agile Development Practices
Group work was structured around sprint-based (iterative short and time-bound) coding activities which focused on real-world genomics issues. The sprints developed real-world programming skills, for example, learning how to collect data from remote application programming interfaces (APIs), filter the data and present only relevant data to the end user.

There were three sprints during the 10 week course with each one increasing in difficulty. The students worked together in teams to support each other whilst playing different roles in the process, such as, Project lead, Programmer and Tester. All best practice methods were incorporated in the course, such as version controlling their group work using GitHub. All the tools and platforms commonly used by Clinical Bioinformaticians were brought into use during the course. All the learning materials were downloaded to the students machines so they could be taken away and referred to when back in practice

3 Pedagogy

The design was from a "Pedagogy-first" approach (prioritising learning methods over tools) which resulted in 3 pedagogical themes:

Social Learning
Group work encouraged peer learning, reduced isolation,& helped focus the facilitation on those who needed it most

Problem Solving
Real-world problems drove the group work & individual activities to develop self-regulated learning skills

Situated Learning
Simulated real-world practices included authentic programming issues to develop the skills & know-how needed back in practice

Created by priyanka from Noun Project
Created by Nithinan Tatah from Noun Project
Created by Vectors Point from Noun Project

4 Tools

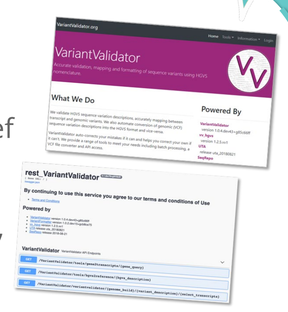
Slack
Provided an informal team-based platform that helped to discuss issues and share files. For example ,over 600 messages & 123 resources were shared

Jupyter Notebooks
interactive web-based workbooks provided editable coding exercises next to teaching materials which helped students to learn to code in a fail-safe environment

GitHub
Students used this industry-preferred platform to download the notebooks & version control the group work projects

5 Back to Practice

VariantValidator
The coding requirements had been driven and designed to meet the brief of a real-world genomics problems. Aspects of the code have been tidied up, documented and integrated into VariantValidator API.



Some students have been invited to work on the VariantValidator project where their code can make a real different in genomic medicine. They will also co-design new materials for the next run of the course in November therefore ensuring the relevance and currency of the course

Links to the course: https://www.manchester.ac.uk/study/masters/courses/list/12099/pgcert-clinical-bioinformatics/

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